

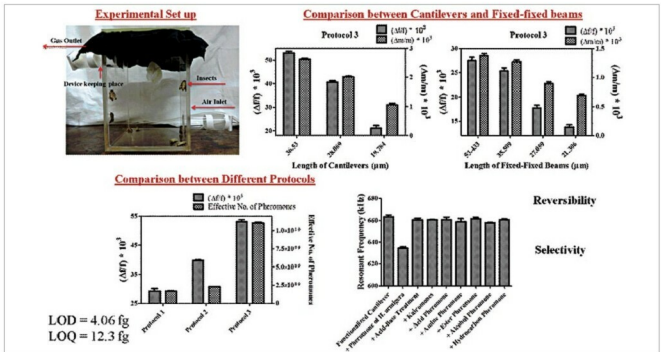


Fig. 4: Selective and reversible sensing of *H. armigera* pheromone

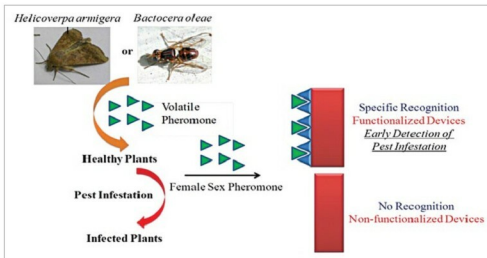


Fig. 5: Summary

To accomplish this target, the scientists fabricated a few MEMS devices (Fig. 1) and then covalently functionalised them for selective sensing of a particular pheromone. Using these devices, they were able to detect the female sex pheromone of *Helicoverpa armigera*, *Scirpophaga incertulas* (Fig. 2) and *Bactrocera oleae* (Fig. 3)—the most known agriculturally hazardous pests globally.

The increase in pheromone mass was quantitatively sensed by a pro-

portionate change in frequency or resistance, which was measured and monitored continuously (Fig. 4). Experiments were also performed with male and female *Helicoverpa armigera* pests in presence of tomato plants in a confined region that mimicked the real environmental conditions.

After selective detection of pheromone and prior to applying any measure to control the pest plague, a stage was identified where the pheromone concentration was just

below the danger level. Necessary actions were then taken as and when needed in a confined region of alerted pest attack. This also reduced the possibility of anthropogenic contamination due to the current overuse of pesticides/insecticides.

These pheromone nanosensors show excellent efficacy and stability in open atmosphere, even during wet atmospheric conditions like rainy season or in peak summer under bright sunshine. This feature heightens their potential for commercial use in any season under ambient conditions. Again, the transportation of these pheromone nanosensors is trouble-free because of covalent bonding.

Overall, this is a novel bioengineering approach for selective and reversible sensing of female sex pheromones of certain hazardous pests, which may be efficiently and economically carried forward from the research laboratory to the agricultural field. **EFY**